Classifying Initial Pleadings in Health-Related Lawsuits: Supervised Model Comparison

Health-Lawsuit Classification Using Supervised Models

MARCELO LISBOA ROCHA*

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, mlisboa@uft.edu.br

ALESSANDRA LIMA COSTA

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, alessandra.lima@uft.edu.br

ÂNGELA CARMEN SZYMCZAK DE CARVALHO

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, angela.carmen@uft.edu.br

FELIPE OLIVEIRA COLEN

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, felipe.colen@uft.edu.br

DAVID NADLER PRATA

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, ddnprata@uft.edu.br

GENTIL VELOSO BARBOSA

Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, gentil@uft.edu.br

Accurate subject classification of initial pleadings can support judicial case management, statistical reporting, and document triage. This study investigates supervised classification of health-related lawsuits from the Court of Justice of the State of Rondônia, Brazil. The source corpus comprises 24,106 lawsuits filed between 2005 and 2025, associated with 27,056 subject records. Initial-pleading extraction succeeded for 24,623 records, and seven frequent subject categories were selected for model development. Only fully adjudicated lawsuits were included. For every retained case, the initial-pleading text was used as the model input, and the corresponding TPU subject label was individually verified by a qualified judicial specialist from the court. The resulting expert-verified labels were treated as the gold-standard reference for the experiments. The experiments combined term frequency–inverse document frequency (TF-IDF) representations, class-balancing strategies, and classifier comparison using PyCaret 3.3.2, alongside separate multilayer perceptron (MLP) and long short-term memory (LSTM) configurations. Within the PyCaret benchmark, LightGBM achieved the highest reported mean F1-score of 0.9154, with accuracy of 0.9152, Cohen's kappa of 0.8983, and a Matthews correlation coefficient of 0.8988. Training-only oversampling yielded higher scores than undersampling under the evaluated protocol. In separate neural experiments, selecting the 50 most discriminative terms increased reported accuracy from approximately 0.87 to 0.99 for both architectures. These results are not directly comparable with the PyCaret ranking and require further robustness assessment. The findings demonstrate the potential of supervised models for classifying health-related initial pleadings. Further evaluation under identical partitions, repeated experimental runs, calibration analysis, and qualitative error analysis is recommended before operational adoption.

Corresponding author: Marcelo Lisboa Rocha, mlisboa@uft.edu.br. Postal address: Quadra 604 Sul, Alameda 12, no. 29, Palmas/TO, Brazil, 77022-032. Telephone: +55 63 98405-5547.

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1 INTRODUCTION

Health-related litigation produces a large and heterogeneous stream of documents whose procedural metadata support case assignment, court statistics, management dashboards, and downstream artificial-intelligence systems. The subject assigned to a lawsuit is not a peripheral attribute: it links the pleading to a formal taxonomy and affects how the institution describes its caseload.

The task examined here is automatic subject classification from the text of an initial pleading. The study uses records from the Electronic Judicial Process system (PJe) of the Court of Justice of the State of Rondônia (TJRO), Brazil. The analysis focuses on seven frequent health-related subject categories and compares supervised classification configurations based on TF-IDF representations.

The study evaluates how accurately the supervised classifiers reproduce the subject labels used for training and testing. Predictive performance is assessed using accuracy, precision, recall, F1-score, Cohen's kappa, and the Matthews correlation coefficient.

Three contributions are offered. First, the article characterizes the distribution of health-related judicial subjects in a real court corpus. Second, it documents the computational environment, data-partitioning procedure, feature representations, and resampling boundaries. Third, it compares supervised classifiers using multiple complementary performance measures.

The main empirical finding is that LightGBM obtained the highest mean F1-score among the classifiers available through the PyCaret comparison API. The model is a promising candidate for decision support, although class-level error analysis, confidence calibration, repeated experiments, audit trails, and prospective evaluation remain necessary before institutional deployment.

2 RELATED WORK

2.1 Legal text classification and legal NLP

Legal documents combine technical vocabulary, long-range dependencies, normative context, and semantically adjacent categories. LexGLUE was proposed as a benchmark for legal language understanding across several tasks and showed that domain-oriented models can outperform general models in specific legal settings [1]. LegalBench extended the evaluation agenda to 162 tasks formulated collaboratively with legal professionals, reinforcing that legal-model assessment must examine reasoning and domain alignment in addition to aggregate accuracy [2].

For judicial-document classification, Sargeant, Izzidien, and Steffek used an LLM and a new taxonomy to classify UK summary-judgment decisions, reporting 87.13% accuracy and an F1-score of 0.87 [3]. Lee compared traditional machine learning, smaller domain-adapted language models, and general LLMs in Korean criminal-precedent classification; domain-adapted models such as KLUE-BERT could outperform larger general-purpose models [4]. Together, these studies show that model size alone does not determine performance: data quality, taxonomy design, domain adaptation, and error interpretation remain decisive.

Beyond legal NLP, recent classification studies illustrate complementary methodological issues relevant to this design. Basharat et al. analyzed how severe class skew can produce misleading accuracy in social-media credibility assessment [26], supporting the use of multiple metrics and explicit imbalance handling. Mathew et al. used a stacked ensemble including LightGBM for multilevel, multiclass troll classification [27], providing a text-classification comparison for ensemble learning. In medical imaging, Mehan combined Inception v3 with neural-network, gradient-boosting, and decision-tree classifiers and evaluated performance using cross-validation, confusion matrices, calibration plots, and ROC analysis [28]. Although these application domains differ from judicial text, they reinforce the need for domain-specific preprocessing and complementary evaluation rather than reliance on accuracy alone.

The present work evaluates supervised classification of initial pleadings according to the subject codes recorded in the judicial dataset. It examines how different classifiers, balancing strategies, and feature-selection configurations perform across the selected health-related categories.

2.2 Human-in-the-loop governance

A judicial classifier should be evaluated as part of a sociotechnical process. Responsible deployment requires versioned models and taxonomies, documented access and retention controls, confidence-based referral to human review, correction mechanisms, and a clear allocation of institutional responsibility. Predictive metrics should be accompanied by prospective monitoring of model performance and institutional use.

The reliability of the proposed pipeline depends on data quality, taxonomy consistency, experimental design, and the requirements of the intended use.

3 METHODOLOGY

The methodological pipeline contains four stages: data acquisition, document processing, and supervised model comparison. Figure 1 summarizes the sequence and the safeguards applied to reduce leakage and overstatement.



Figure 1: Research pipeline and evaluation safeguards.

3.1 Dataset and acquisition

The source was the PostgreSQL database of the PJe system used by TJRO. The observation window covered lawsuits filed from 1 January 2005 through 31 December 2025. A SQL query required at least one of

30 subject codes related directly or indirectly to health: 9995, 10069, 10244, 10263, 10356, 10434, 10440, 10503, 10892, 11854, 11883, 11884, 11993, 11994, 12223, 12480, 12483, 12484, 12487, 12489, 12491, 12498, 12507, 12508, 12510, 12773, 14690, 14691, 14759, and 14760.

Only fully adjudicated lawsuits—that is, cases that had reached a final judgment—were eligible for inclusion. For each eligible case, the extracted initial pleading was used as the model input, while the corresponding TPU subject code was used as the target label. Every retained target label was individually reviewed by at least one qualified judicial specialist from TJRO, who verified that the assigned TPU subject was consistent with the substantive request expressed in the initial pleading and the completed court record. Only labels confirmed through this item-level review were retained. The TPU provides the standardized institutional taxonomy used for subject classification [23], while TJRO regulations provide for the correction of electronic case registrations when a material error or omission is identified [24]. Because this verification covered every retained target label, the resulting corpus is treated as an expert-verified institutional gold-standard reference set for the purposes of this study. This validation procedure also reduces the risk that the reported model performance reflects label noise rather than substantive classification ability [5]. No LLM-generated or LLM-assisted label was used as an experimental target in the study reported here.

The query returned 27,056 subject records associated with 24,106 distinct lawsuits because one case may contain multiple subjects. SQLAlchemy and pandas read_sql_query were used to construct the dataframe for subsequent processing. Table 1 reports the 22 subject categories actually observed in the result. Near-duplicate descriptions occur because similar wording can belong to different branches of the Brazilian Unified Procedural Tables (TPU).

Table 1: Original judicial-subject distribution (27,056 records)

TPU code	English description	Records	Share (%)
10244	Health Assistance	7,716	28.52
11884	Supply of Medications	3,252	12.02
11883	Medical-Hospital Treatment	2,909	10.75
10434	Health Services	2,750	10.16
10263	Treatment of Own Health	1,843	6.81
12489	Medical-hospital treatment	1,484	5.48
10440	Health Services	1,236	4.57
12487	Supply of medications	1,215	4.49
10069	Medical-Hospital Treatment and/or Supply of Medications	1,108	4.10
9995	Health Services	827	3.06
12491	Medical-hospital treatment	708	2.62
10503	Health Services	597	2.21
10892	Medication/Treatment/Surgery of Unproven Effectiveness	595	2.20
10356	Medical-Hospital Assistance	390	1.44
12508	Compulsory hospitalization	173	0.64
12498	Wheelchair/bath chair/hospital bed	111	0.41

TPU code	English description	Records	Share (%)
11854	Mental Health	54	0.20
14760	Home Care	40	0.15
12510	Voluntary hospitalization	25	0.09
12484	Supply of medications	20	0.07
14690	Emergency medical-hospital attendance condition	2	0.01
12483	Hospital Admission/Transfer	1	0.00

The annual counts in Figure 2 show a relatively stable series between 2015 and 2019, a marked reduction in 2020 and 2021, and renewed growth from 2022 onward. Because the design is observational, these changes are described without attributing causality.

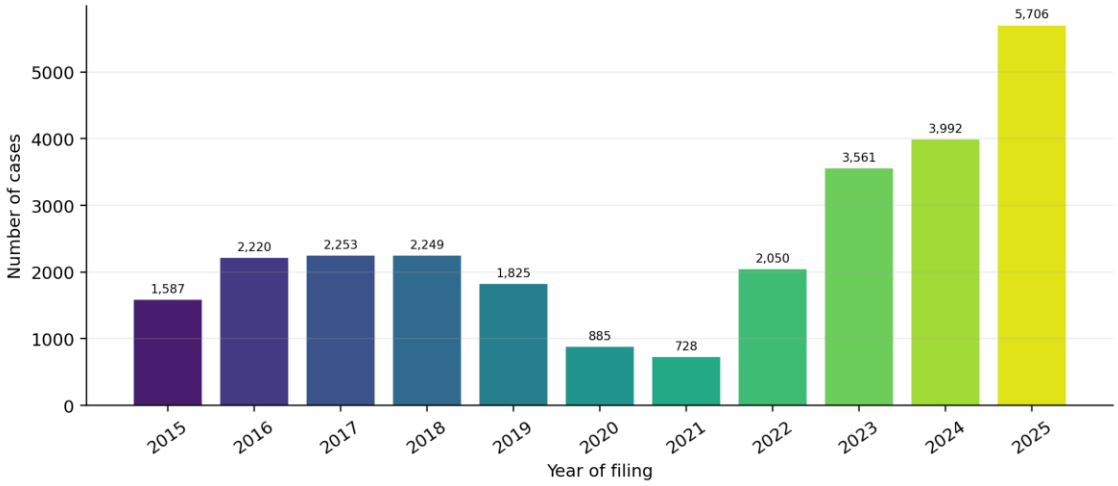


Figure 2: Distribution of lawsuits by filing year, 2015–2025.

3.2 Initial-pleading extraction and preprocessing

The initial pleading was identified and its text extracted through the Codex Platform of the Brazilian National Council of Justice, which provides document-identification and text-extraction services. Extraction succeeded for 24,623 of the 27,056 subject records. In the remaining 2,433 records, the service did not automatically identify the initial pleading or return its text, that were excluded. Consequently, these 24,623 records constitute the initial dataset used in this study.

After, a Pareto analysis was applied to identify the classes with sufficient frequency for training. Seven of the 22 subjects accounted for 78% of the records, as shown in Figure 3 and Table 2.

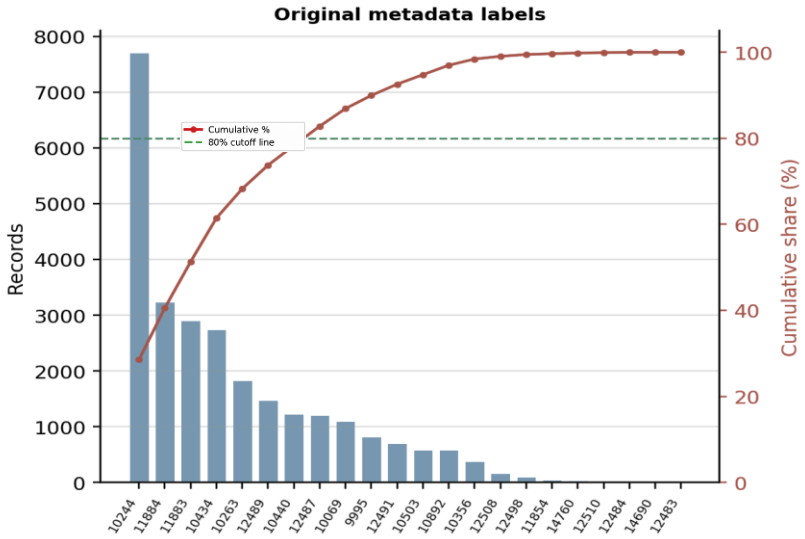


Figure 3: Original distribution of metadata labels using Pareto analysis.

Table 2: Relevant subject labels according to Pareto analysis

TPU code	Description	Total	%	Cumulative %
10244	Health Care	7,716	28.52%	28.52%
11884	Provision of Medications	3,252	12.02%	40.54%
11883	Medical-Hospital Treatment	2,909	10.75%	51.29%
10434	Health Services	2,750	10.16%	61.45%
10263	Treatment of One's Own Health	1,843	6.81%	68.27%
12489	Medical-Hospital Treatment	1,484	5.48%	73.75%
10440	Health Services	1,236	4.57%	78.32%

As shown in Figure 3, Pareto analysis identified categories with enough observations for model development. Seven of the 22 original subjects accounted for approximately 78% of the records. The corpus was also processed with a domain-specific stopword inventory: 132 legal terms, 28 common personal-data references, 12 ordinary high-frequency words, 10 number/value expressions, 11 administrative terms, 51 lexical variants, and 195 additional items. Removal was applied cautiously because apparently generic legal expressions can still discriminate between adjacent procedural categories. This way, the final dataset, as stated in Table 2, counts with 21,190 records.

3.3 Text representation, class balancing, and models

The training dataframe contained two variables: text, representing the extracted initial-pleading text used as the model input, and cd_assunto, representing the individually verified TPU subject code used as the gold-standard target label. Texts were represented with term frequency–inverse document frequency (TF-IDF), which weights a term by its within-document frequency and its rarity across the corpus [6]. TF-IDF emphasizes discriminative vocabulary but does not preserve word order or long contextual dependencies [9].

$$\text{TF–IDF}(t,d,D) = \text{TF}(t,d) \times \text{IDF}(t,D), \quad \text{IDF}(t,D) = \log(|D| \,/\, |\{d \in D : t \in d\}|)$$

Class imbalance was addressed with both undersampling and oversampling. Undersampling lowers computational cost by removing majority observations but may discard useful boundary cases. Oversampling increases minority representation; techniques such as SMOTE construct synthetic points from neighboring examples, although interpolated vectors in a text-feature space are not necessarily equivalent to legally plausible new pleadings. Comparative studies show that the preferred strategy depends on the data and must be evaluated outside the resampled training set [16, 17, 18, 19].

Light Gradient Boosting Machine (LightGBM) is a tree-based gradient-boosting implementation designed for efficient and scalable training [7]. Its performance cannot be assumed for a new corpus, so the study compared it with all classification methods made available through the PyCaret API under the same automated protocol. Prior domain applications demonstrate the need for empirical, rather than universal, claims of superiority [8].

Two neural configurations were also retained. A multilayer perceptron (MLP) received sparse TF-IDF vectors and learned nonlinear relations between input features and output classes. The approach is comparatively inexpensive and remains useful when the vocabulary itself must be inspected [10, 11]. A Long Short-Term Memory (LSTM) network was also tested with TF-IDF input [12, 13, 14]. This combination must be interpreted cautiously because an LSTM was designed for sequences whereas TF-IDF aggregates counts and removes word order. It therefore behaves here as a neural classifier over static features and does not fully exploit temporal modeling. A proper sequential comparison would require ordered tokens or contextual embeddings [15].

3.4 Computational environment and experimental protocol

Experiments ran in Python 3.11. Automated classifier comparison used PyCaret 3.3.2. The study records the language version, API version to reduce ambiguity in reproduction. Six focal configurations were evaluated, as summarized in Table 3.

Table 3: Focal experimental configurations

ID	Classifier	Size	Balancing	Features
C1	LightGBM	100/class	Undersampling	All TF-IDF features
C2	LightGBM	500/class	Training-only oversampling	All TF-IDF features
C3	MLP	500/class	Balanced training	All TF-IDF features
C4	LSTM over TF-IDF	500/class	Balanced training	All TF-IDF features
C5	MLP	500/class	Balanced training	Top 50 discriminative terms
C6	LSTM over TF-IDF	500/class	Balanced training	Top 50 discriminative terms

Data splitting was performed at the lawsuit level so that text associated with the same case could not appear in both training and test sets. Oversampling was restricted to the training data, while validation and test sets retained real observations and their original class distributions. The selection of the 50 most discriminative words was performed inside each validation fold without access to the test set. These boundaries are intended to reduce leakage; they do not resolve uncertainty in the substantive correctness of the targets. For the neural configurations, the balanced training set contained 3,500 pleadings (500 per class),

and the held-out test set contained 700 pleadings (100 per class). For the LightGBM sampling comparison, undersampling used 700 observations (100 per class), whereas training-only oversampling balanced the training data to 3,500 observations (500 per class). The archived outputs do not retain a distinct validation-set count or a single identical partition across the PyCaret and neural experiments; the experiments are therefore reported separately.

3.5 Evaluation measures and interpretation rules

The primary ranking measure was the mean F1-score reported by PyCaret. F1 is the harmonic mean of precision and recall and is useful when both false-positive and false-negative assignments matter. Accuracy was retained because it gives the overall proportion of correctly reproduced targets, but accuracy alone can be optimistic in an imbalanced multiclass corpus. Precision describes how often a predicted subject is the recorded target, whereas recall describes how often records of that target are recovered. The reported values are aggregate cross-validation summaries; they do not replace classwise inspection.

AUC, Cohen's Kappa, and the Matthews correlation coefficient (MCC) were treated as complementary diagnostics. AUC summarizes ranking discrimination when the implementation exposes a compatible probability or decision score. Kappa compares observed agreement with agreement expected by chance. MCC incorporates all cells of a confusion matrix and is informative when classes are unevenly represented. The combination of F1, Kappa, and MCC was therefore used to check whether a high accuracy value merely reflected concentration in dominant subjects.

The comparison used the cross-validation means supplied by PyCaret experiment. Standard deviations were available for the LightGBM sampling comparison and are reported in Table 8. No formal paired significance test or confidence interval for the difference between algorithms was available in the source experiment. Consequently, a first-place rank is interpreted as an empirical ordering under the recorded folds, not as proof that LightGBM will dominate every alternative in a new sample. The 0.0076 F1 difference between LightGBM and Gradient Boosting is reported so that the practical size of the margin remains visible.

The evaluation uses accuracy, precision, recall, F1-score, AUC, Cohen’s kappa, and MCC to examine performance from complementary perspectives. All models were evaluated against the expert-verified gold-standard labels described in the dataset-acquisition protocol. Accordingly, the reported scores measure agreement with target labels that underwent item-level verification by qualified TJRO judicial personnel, rather than with unreviewed administrative metadata.

4 RESULTS

4.1 Comparison of the Classifiers Available in PyCaret

The compare_models function in PyCaret 3.3.2 was executed to compare the classifiers available through the API under the same procedure. The complete ordering reported in Table 9 uses mean F1-score.

LightGBM ranked first, with F1-score of 0.9154, accuracy of 0.9152, precision of 0.9181, and recall of 0.9152. It also achieved Kappa of 0.8983 and MCC of 0.8988, which are compatible with high agreement beyond chance in the evaluated dataset. Its reported AUC was 0.9897. Because Table 9 contains AUC values of 0.0000 for several methods, that column should not support comparisons across all models without further clarification of metric availability or calculation for each estimator.

The advantage over the second-ranked Gradient Boosting Classifier was 0.0076 in F1-score, equivalent to 0.76 percentage point. This result supports the selection of LightGBM within the PyCaret comparison, although the narrow difference recommends examining stability across folds, inference cost, and class-level behavior. LightGBM's mean training time was 5.11 seconds, longer than that of several faster competitors.

The metrics in Table 9 should not be combined directly with the MLP and LSTM results obtained with Top-50 feature selection because these configurations belong to separate experiments. A conclusive comparison requires the same test set, folds, sampling unit, and preprocessing for every model.

In the TF-IDF + MLP configuration without Top-50 selection, the detailed classification report was as follows.

Table 4: Detailed classification report for TF-IDF plus MLP without Top-50 feature selection

Class	Precision	Recall	F1	Support
11884	0.99	1.00	1.00	100
10244	1.00	1.00	1.00	100
10263	0.65	0.80	0.72	100
10440	0.87	0.87	0.87	100
12489	0.89	0.85	0.87	100
11883	0.99	1.00	1.00	100
10434	0.71	0.57	0.63	100
Accuracy			0.87	700
Macro avg	0.87	0.87	0.87	700
Weighted avg	0.87	0.87	0.87	700

Table 4 shows the baseline MLP results without feature selection. The aggregate accuracy, macro-average F1-score, and weighted-average F1-score were all 0.87 across the 700 evaluated observations; the lower F1-scores for classes 10263 and 10434 identify the principal class-level limitations of this configuration.

In the TF-IDF + LSTM configuration without Top-50 selection, the detailed classification report was as follows.

Table 5: Detailed classification report for TF-IDF plus LSTM without Top-50 feature selection

Class	Precision	Recall	F1	Support
11884	1.00	1.00	1.00	100
10244	1.00	1.00	1.00	100
10263	0.79	0.60	0.68	100
10440	0.90	0.89	0.89	100
12489	0.89	0.79	0.84	100
11883	0.99	1.00	1.00	100
10434	0.60	0.81	0.69	100
Accuracy			0.87	700
Macro avg	0.88	0.87	0.87	700
Weighted avg	0.88	0.87	0.87	700

Table 5 reports the corresponding LSTM baseline. Its aggregate accuracy and macro- and weighted-average F1-scores were also 0.87, while classes 10263 and 10434 remained the least accurately classified, indicating that the TF-IDF representation did not eliminate the most difficult class distinctions.

Classes 10263 (Hospital Admission/Transfer) and 10434 (Medical-hospital treatment) concentrated the lowest F1-scores in both neural networks. Their semantic proximity to other health-related subjects offers a plausible explanation for some of the confusion, but this hypothesis requires qualitative review of the misclassified cases. The archived outputs did not retain case-level predictions, so a document-level qualitative error analysis could not be reconstructed; interpretation is therefore limited to the observable class-level pattern, and case-level review is identified as future work.

Selecting the 50 most discriminative words reduced dimensionality and removed weakly informative features. Before this refinement, the MLP achieved F1-score of 0.72 for class 10263 and 0.63 for class 10434; the corresponding LSTM scores were 0.68 and 0.69.

After Top-50 selection, both configurations reached aggregate accuracy of 0.99 on the evaluated set. A gain of this magnitude requires external testing and repetition with different random seeds because it may reflect strong vocabulary separability, partition sensitivity, or overfitting to the selection procedure. Even with feature selection performed within the folds, this result should be verified through repeated runs, confidence intervals, and testing on an external dataset [20, 21, 22].

Table 6: Detailed classification report for TF-IDF plus MLP with Top-50 feature selection

Class	Precision	Recall	F1	Support
11884	1.00	1.00	1.00	100
10244	1.00	1.00	1.00	100
10263	1.00	0.95	0.97	100
10440	1.00	1.00	1.00	100
12489	1.00	1.00	1.00	100
11883	1.00	1.00	1.00	100
10434	0.95	1.00	0.98	100
Accuracy			0.99	700
Macro avg	0.99	0.99	0.99	700
Weighted avg	0.99	0.99	0.99	700

Table 6 shows the MLP after Top-50 feature selection. Aggregate accuracy and the macro- and weighted-average F1-scores increased to 0.99; the remaining departures from perfect class-level scores were concentrated in classes 10263 and 10434.

Table 7: Detailed classification report for TF-IDF plus LSTM with Top-50 feature selection

Class	Precision	Recall	F1	Support
11884	1.00	1.00	1.00	100
10244	1.00	1.00	1.00	100
10263	1.00	0.95	0.97	100
10440	1.00	1.00	1.00	100
12489	1.00	1.00	1.00	100
11883	1.00	1.00	1.00	100
10434	0.95	1.00	0.98	100
Accuracy			0.99	700
Macro avg	0.99	0.99	0.99	700
Weighted avg	0.99	0.99	0.99	700

Table 7 presents the LSTM after the same Top-50 feature selection. It achieved aggregate accuracy and macro- and weighted-average F1-scores of 0.99, supporting a strong result in this evaluated split while retaining the need for repeated and external validation.

4.2 LightGBM under sampling strategies

With undersampling, majority classes were reduced to 100 examples. With training-only oversampling, the LightGBM model produced higher mean values under every reported metric. Table 8 reports the cross-validation means and standard deviations. The result favors oversampling within this experiment, but it does not establish that synthetic sampling is universally preferable for judicial text.

Table 8: LightGBM metrics by training-set sampling strategy

Strategy	Stat.	Acc.	AUC	Recall	Prec.	F1	Kappa	MCC
Undersampling	Mean	.8238	.9661	.8238	.8302	.8201	.7886	.7919
Undersampling	Std	.0372	.0113	.0372	.0415	.0401	.0446	.0442
Oversampling	Mean	.9152	.9897	.9152	.9181	.9154	.8983	.8988
Oversampling	Std	.0188	.0034	.0188	.0199	.0187	.0225	.0228

4.3 Comparison of PyCaret classifiers

PyCaret compare_models were executed to compare the classifiers exposed by the API under the same procedure. Table 9 is ordered by mean F1-score. LightGBM ranked first with F1 = 0.9154, accuracy = 0.9152, precision = 0.9181, recall = 0.9152, Kappa = 0.8983, and MCC = 0.8988. Its AUC was 0.9897.

The F1 advantage over the second-ranked Gradient Boosting Classifier was 0.0076, or 0.76 percentage point. The margin supports selecting LightGBM within this benchmark but is small enough to justify examining fold stability, inference cost, and per-class behavior. LightGBM's mean training time was 5.11 seconds, slower than several alternatives. Values of AUC = 0.0000 for some estimators should not be interpreted as genuine zero discrimination without clarification of metric availability or calculation for those models.

Table 9: PyCaret classifier comparison ordered by mean F1-score

Model	Acc.	AUC	Recall	Prec.	F1	Kappa	MCC	Sec.
LightGBM	.9152	.9897	.9152	.9181	.9154	.8983	.8988	5.110
Gradient Boosting	.9076	0.0000	.9076	.9099	.9078	.8891	.8896	44.906
Ridge	.9048	0.0000	.9048	.9081	.9045	.8857	.8866	.956
Linear SVM	.9038	0.0000	.9038	.9081	.9035	.8846	.8857	1.215
Extra Trees	.9024	.9841	.9024	.9035	.9015	.8829	.8834	.768
Random Forest	.9005	.9842	.9005	.9020	.8999	.8806	.8812	.616
Decision Tree	.8919	.9350	.8919	.8922	.8907	.8703	.8708	.670
Logistic Regression	.8905	0.0000	.8905	.8933	.8900	.8686	.8694	1.854
Naive Bayes	.8562	.9137	.8562	.8601	.8538	.8274	.8292	.445
LDA	.8333	0.0000	.8333	.8490	.8324	.8000	.8036	6.825
K Neighbors	.8014	.9478	.8014	.8064	.7964	.7617	.7653	.661
QDA	.7710	0.0000	.7710	.8128	.7815	.7251	.7301	3.416
AdaBoost	.3367	0.0000	.3367	.2835	.2288	.2040	.3494	2.535
Dummy	.1667	.5000	.1667	.0278	.0476	0.000	0.000	.404

4.4 Neural configurations and Top-50 selection

Without Top-50 selection, both MLP and LSTM achieved aggregate accuracy of 0.87. The lowest class F1-scores occurred for code 12483 (Hospital Admission/Transfer) and code 12491 (Medical-hospital treatment). The semantic proximity of these categories to other health subjects is a plausible explanation, but qualitative review of misclassified pleadings is required before accepting that interpretation.

For the MLP, F1-scores were 0.72 for class 12483 and 0.63 for class 12491; for the LSTM they were 0.68 and 0.69. After selection of the 50 most discriminative terms, both neural configurations reported aggregate accuracy of 0.99, with class F1-scores of 0.97 and 0.98 for those two difficult classes. Figure 4 displays the classwise differences.

A jump of this magnitude requires external testing and repetition with different random seeds. It may reflect genuinely separable vocabulary, sensitivity to the partition, or overfitting to the feature-selection procedure. Although feature selection was performed within each fold, the result should be verified through repeated runs, confidence intervals, and external testing. Reports of large gains from feature-selection and hybrid neural pipelines in other application domains provide useful methodological comparisons [20, 21, 22].

The Top-50 neural results must not be combined directly with the PyCaret ranking. They belong to different experimental configurations. A conclusive architecture comparison would require the same test set, folds, sampling unit, resampling policy, feature-selection procedure, and preprocessing for every model.

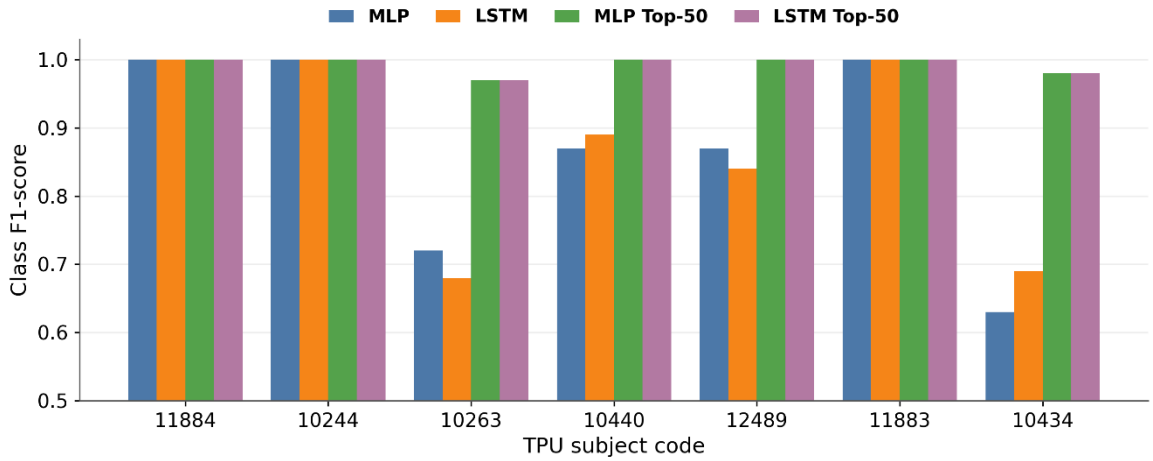


Figure 4: Classwise F1-scores for the neural TF-IDF configurations.

5 DISCUSSION

5.1 Interpreting the LightGBM result

LightGBM was the strongest model under the PyCaret comparison: F1 = 0.9154 with Kappa and MCC near 0.90. These complementary metrics reduce the likelihood that the ranking is an artifact of majority-class accuracy alone. Oversampling also outperformed undersampling under the recorded protocol. The result is technically encouraging because it combines strong discrimination with moderate training time.

Nevertheless, the difference between LightGBM and Gradient Boosting was narrow. Model selection for court use should therefore consider more than the first row of an automated leaderboard. Relevant criteria include class-level recall for rare subjects, probability calibration, robustness to taxonomy revisions, inference latency, memory consumption, interpretability, and the cost of sending uncertain cases to human review.

5.2 Institutional use and governance

The safest near-term use is decision support. The classifier can recommend one or more candidate subjects, display confidence, and route low-confidence or high-impact cases to trained court personnel. Every recommendation should retain the model version, taxonomy version, input-document identifier, output ranking, and human decision. These records make later correction and audit possible.

The use of unredacted pleadings in a reserved environment also creates governance duties. Contractual isolation is not equivalent to a complete privacy and security program. Operational deployment should define lawful purpose, minimum necessary data, access profiles, retention periods, logging, incident response, and whether the model provider may use prompts or outputs for service improvement. These controls should be reviewed before scaling beyond a research proof of concept.

5.3 Proposed operational evaluation protocol

Before operational use, the selected model should be evaluated on an external test set representing different filing periods, subject frequencies, and document characteristics. The evaluation should include per-class precision, recall, F1-score, confusion matrices, probability calibration, processing time, and qualitative examination of classification errors. Special attention should be given to rare categories and semantically similar subjects that produced lower F1-scores in the neural experiments.

A deployment pilot should operate in shadow or recommendation mode. The system would rank candidate subjects and record confidence without directly replacing official metadata. Trained personnel would accept, reject, or amend each recommendation. Review thresholds could be set conservatively, with mandatory human handling for low-confidence predictions, rare subjects, novel vocabulary, and cases whose consequences for reporting or workflow are high. Monitoring should track coverage, abstention, per-class precision and recall, calibration, reviewer overrides, processing time, and drift in document language or taxonomy use.

Version control is essential because both the classifier and the judicial taxonomy can change. Each prediction record should identify the data-preprocessing version, feature vocabulary, model and hyperparameters, training interval, taxonomy files, and confidence output. When the taxonomy changes, historical comparability must be assessed before retraining or remapping labels. A rollback mechanism and an auditable correction process are also required. These steps convert the proof of concept into a testable institutional workflow while preserving the study's central constraint: automated consistency cannot substitute for legal responsibility.

5.4 Limitations and validity threats

The first limitation is geographic and institutional: the corpus comes from one Brazilian state court and one family of health-related matters. Vocabulary, document templates, filing practices, and subject distributions may differ in other courts or periods. The second concerns selection: text extraction failed for 2,433 records. Although every retained label was individually verified by at least one qualified judicial specialist, the study did not employ independent double annotation, adjudication of disagreements, or inter-annotator agreement. Accordingly, the term gold standard refers only to an item-level, expert-verified institutional reference set within TJRO and should not be interpreted as a multi-annotator consensus benchmark [25].

A further limitation is comparability: the PyCaret benchmark and separate MLP/LSTM Top-50 experiments should not be interpreted as a single unified leaderboard.

Future work should preserve a temporal external test set, repeat experiments with multiple random seeds, evaluate confidence calibration, compare all architectures under identical partitions, and assess model drift after changes in taxonomy or filing practices.

6 CONCLUSION

This study investigated automatic subject classification of initial pleadings in health-related lawsuits from a Brazilian state court. The experimental pipeline combined text extraction, selection of seven frequent subject categories, TF-IDF representation, class balancing, and supervised model comparison. Its principal contribution is an empirical comparison of supervised models, balancing strategies, and feature-selection configurations for classifying health-related initial pleadings.

Restricting the corpus to fully adjudicated lawsuits and individually verifying every target label provides a stronger reference basis for the supervised experiments. In this study, the resulting labels are therefore regarded as an expert-verified institutional gold standard: the initial-pleading text constitutes the model input, and the judicially verified TPU subject constitutes the reference target.

Within the PyCaret benchmark, LightGBM achieved the highest reported mean F1-score of 0.9154, with accuracy of 0.9152, Cohen's kappa of 0.8983, and a Matthews correlation coefficient of 0.8988. Training-only oversampling outperformed undersampling under the reported protocol. However, the F1-score difference of 0.0076 between LightGBM and Gradient Boosting does not establish a reliable performance advantage without further uncertainty analysis. LightGBM should therefore be regarded as a candidate for further evaluation rather than a universally superior classifier.

The separate MLP and LSTM experiments reported an increase in accuracy from approximately 0.87 to 0.99 after selection of the 50 most discriminative terms. This finding warrants investigation through repeated runs, verification of feature-selection and resampling boundaries, duplicate-document checks, and qualitative examination of correct and incorrect predictions. Because the neural experiments and the PyCaret benchmark were conducted under different configurations, their results do not support a unified ranking of the architectures.

The study remains limited by its single-court setting, restriction to frequent health-related subjects, and exclusion of records for which text extraction failed. Further work should report exact partition sizes, compare all models using identical partitions, and test the selected configurations on data from later periods and other courts. Evaluation should include macro and per-class metrics, repeated-seed results or confidence intervals, probability calibration, and qualitative analysis of classification errors.

The proposed pipeline is best understood as a proof of concept for assisting judicial personnel with subject classification. Any operational pilot should preserve human control over final assignments, document model and taxonomy versions, and maintain appropriate privacy safeguards and audit trails. These measures would allow its potential benefits for consistency, case management, and document triage to be evaluated in practice.

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